UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026).
Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call.
Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	FALSE	A vector image becomes pixelated when enlarged.	Vector images scale without pixelation.
A2	FALSE	Interlacing displays every line of a video frame at the same time.	Interlacing draws alternate lines.
A3	TRUE	Multimedia combines text, graphics, audio, video and animation.	These are the core multimedia elements.
A4	FALSE	Anti-aliasing makes the edges of text look more jagged.	Anti-aliasing smooths jagged edges.
A5	FALSE	A plotter is an input device.	A plotter produces output.
A6	FALSE	RAM is a type of permanent storage.	RAM is temporary working memory.
A7	FALSE	A linear presentation allows users to choose their own path.	A linear presentation follows a fixed sequence.
A8	TRUE	Path animation moves an object along a predefined route.	Path animation follows a motion path.
A9	FALSE	Interactivity level is irrelevant when choosing an authoring tool.	The required interactivity affects tool choice.
A10	TRUE	Frame rate is the number of video frames displayed per second.	Frame rate is measured in fps.
A11	TRUE	A scanner is an input device.	A scanner captures documents into the computer.
A12	TRUE	RGB is an additive colour model.	RGB adds red, green and blue light.
A13	TRUE	Tweening creates intermediate frames between keyframes.	Tweening fills the gaps.
A14	TRUE	CMYK is used for printed output.	CMYK is a subtractive print model.
A15	TRUE	A URL identifies the address of a web resource.	A URL is a resource locator.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	G	Storyboard	visual sequence plan
B2	C	Keyframe	important frame in an animation
B3	A	Multimedia	integration of text, graphics, audio, video and animation
B4	M	Tweening	generating in-between frames
B5	E	Copyright	exclusive rights of a creator
B6	K	Linear multimedia	presentation without user control of sequence
B7	N	Cel animation	hand-drawn frame-by-frame animation
B8	O	Scanner	input device that digitises documents
B9	F	Card-based tool	authoring tool organised like pages or cards
B10	I	Sampling	converting sound into digital values
B11	J	Bitmap	image made of pixels
B12	D	Prototype	early model of a product
B13	L	Browser	software that displays web pages
B14	H	MIDI	standard for musical notes and instructions
B15	B	Hypermedia	linked multimedia elements

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	C	Musical notes and instructions
C2	D	Analog sound to digital values
C3	A	Modelling and rendering
C4	D	Structure web content
C5	B	Printed output
C6	D	Pixels
C7	A	Draws alternate lines of a frame
C8	C	Card-based
C9	C	WAV
C10	A	Assemble multimedia elements
C11	C	Space between individual character pairs
C12	C	Compiler
C13	D	Small strokes at character ends
C14	A	It combines visuals and sound
C15	D	Replace a colour background
C16	A	A visual sequence plan
C17	D	Text, graphics, audio, video and animation
C18	B	Vector
C19	B	Improves accuracy
C20	C	Drawing each frame by hand
C21	D	Multimedia with hyperlinks
C22	A	Temporary working memory
C23	C	A simulated environment
C24	C	Number of pixels in an image
C25	A	A cinema presentation
C26	D	Display web pages
C27	C	Another resource
C28	D	Drawing input
C29	C	A video compression family
C30	C	A low-fidelity layout sketch

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PART D - WORKED ANSWERS

QUESTION 1

Topic: text

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

Sans serif is cleaner at screen sizes; serif aids reading in print.

(2 marks)

b) Describe TWO roles of hypertext in multimedia.

It links related content and supports non-linear navigation.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

Clear headings and links guide users through HTML pages and make the information easier to scan and follow.

(5 marks)

QUESTION 2

Topic: design project

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

A project manager and an interface designer are important roles.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

A simple sketch with a title, logo and large topic buttons is sufficient at low fidelity.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

Portability and constant connectivity make mobile devices convenient for games and shopping.

(4 marks)

QUESTION 3

Topic: graphics

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

Bitmap images use pixels and can pixelate; vector images use mathematical objects and scale cleanly.

(3 marks)

b) Describe TWO factors that affect image quality.

Resolution and colour depth both affect how much detail an image can show.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

Screens use RGB and print uses CMYK; print normally needs a higher resolution for sharp output.

(4 marks)

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QUESTION 4

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(4 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(4 marks)

END OF MARKING SCHEME

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